

Filling in the Blanks: Addressing Positivity Violations with Mathematical Models

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¹Footnotes are for asides or references

Motivating Example

Question: What is the mean systolic blood pressure (SBP) in mm Hg among children and adolescents aged 2-17 in the United States?

Data: NHANES 2017-2018

Challenge: SBP is missing for 44% children

Notation

Y : systolic blood pressure (SBP)

R : indicator if SBP was measured, with 1 being yes

X : a vector of covariates

$E[\cdot]$: expected value function

$I(\cdot)$: indicator function, equal to 1 when true, 0 otherwise

Parameter: mean SBP if there was no missing data

$$\mu = E[Y]$$

Complete-Case Analysis

A complete-case analysis assumes

$$E[Y] = E[Y \mid R = 1]$$

- Missingness is independent of SBP (MCAR)
- Marginal exchangeability



With extent of missingness, results are sensitive to assumption

Addressing Missing Data

Exchangeability: SBP is missing at random conditional on age

$$E[Y \mid X = x] = E[Y \mid X = x, R = 1] \text{ for all values of } x$$



Under this assumption²

$$E[Y] = \sum_{x \in \mathcal{X}} E[Y \mid X = x, R = 1] \Pr(X = x)$$

weighted average of the complete-case means within strata of X

²Estimate with direct maximum likelihood estimator, g-computation

For the previous exchangeability assumption, we need

$$\Pr(R = 1 \mid X = x) > 0 \text{ for all values of } x$$

Every unique age has a non-zero probability of having Y observed

- In the population
- Deterministic or structural positivity

In NHANES 2017-2018,

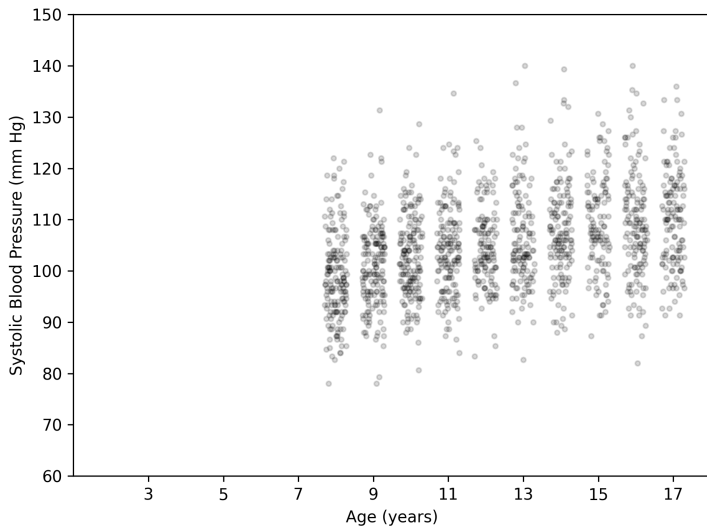
- “BP is measured on participants 8 years and older”³

So $\Pr(R = 1 \mid X = x) = 0$ for $x \in \{2, 3, 4, 5, 6, 7\}$

- *By design* of NHANES

³https://wwwn.cdc.gov/Nchs/Nhanes/2017-2018/BPX_J.htm

NHANES Data



Options for Nonpositivity

1. Change parameter and question
2. Extrapolate from a statistical model
3. Synthesis of statistical and mathematical models

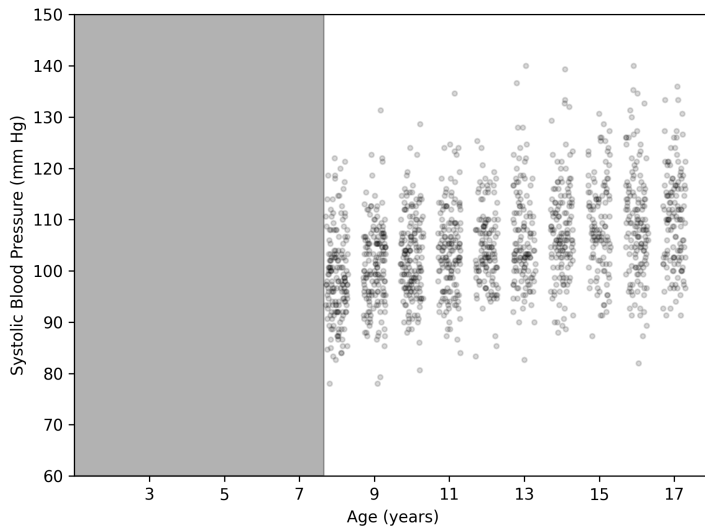
1. Change Parameter

Parameter (revised): mean SBP for those 8 or older

$$E[Y \mid X \geq 8]$$

Question: what is the mean systolic blood pressure (SBP) in mm Hg among children and adolescents **aged 8-17** in the US?

NHANES Data – Restrict



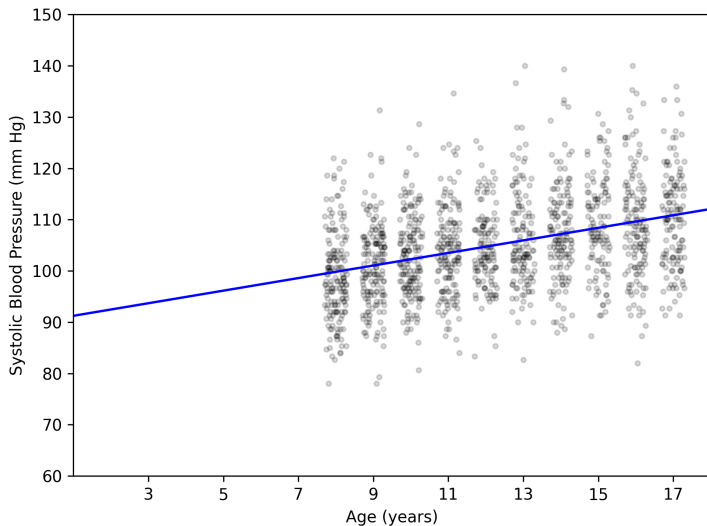
2. Extrapolate

Fit a parametric model

$$E[Y \mid X \geq 8, R = 1] = \beta_0 + \beta_1 X$$

then back-fill missing SBP with model predictions

NHANES Data – Extrapolate



2. Extrapolate

Use of this model has two assumptions:

- a. Linear relationship with age for those 8-17
- b. Linear relationship extends to those 2-7

(a) is in-principle checkable

(b) is not checkable under nonpositivity

3. Synthesis Modeling

Instead of extrapolating, integrate external information

- Address motivating question
- Avoid restrictive assumptions

Divide parameter of interest in parts

The diagram illustrates the decomposition of the overall mean $E[Y]$ into two conditional means based on the sign of X . The equation is $E[Y] = E[Y | X \geq 8] \Pr(X \geq 8) + E[Y | X < 8] \Pr(X < 8)$. The term $E[Y]$ is highlighted in a grey box and labeled "Overall mean" with a green arrow. The term $E[Y | X \geq 8]$ is highlighted in a blue box and labeled "Mean in positive region" with a blue arrow. The term $E[Y | X < 8]$ is highlighted in a red box and labeled "Mean in nonpositive region" with a red arrow.

$$E[Y] = E[Y | X \geq 8] \Pr(X \geq 8) + E[Y | X < 8] \Pr(X < 8)$$

Identification

$$E[Y \mid X \geq 8] = E[E(Y \mid X, R = 1, X \geq 8) \mid X \geq 8]$$

Use a statistical model: $E[Y \mid X, R = 1, X \geq 8]$.⁴

- Estimate using the observed data
- Apply g-computation

⁴The identification result is also why 1. Change Parameter works

Mathematical Model

Identification

$$E[Y \mid X < 8] = ?$$

- Data provide no information

Appeal to external information

- Bounds for continuous Y : $(-\infty, \infty)$
- Bounds for SBP: $[0, 370]$

Assume that⁵

$$E[Y \mid X = x] \in \mathcal{M}_{\alpha, x} \text{ for all } 2 \leq x < 8$$

⁵Structure and variations in mathematical model are rich enough

Mathematical Model

We can do better still

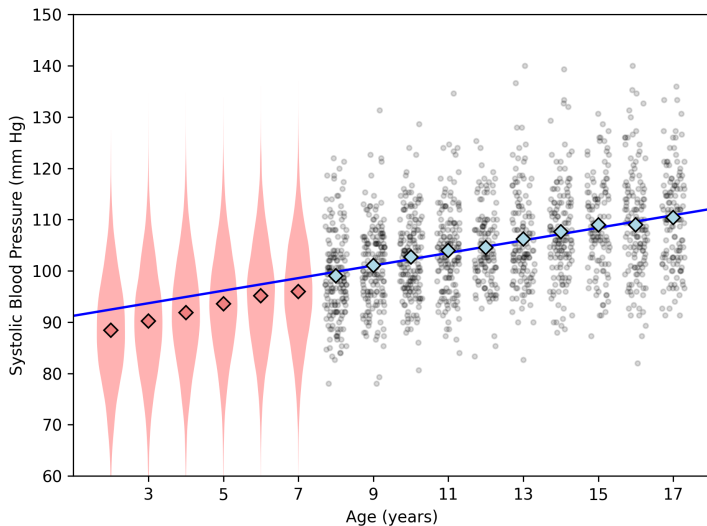
- Published distributions of SBP⁶
- 'Fill-in' missing SBP for those 2-7

Estimation: Monte Carlo Procedure

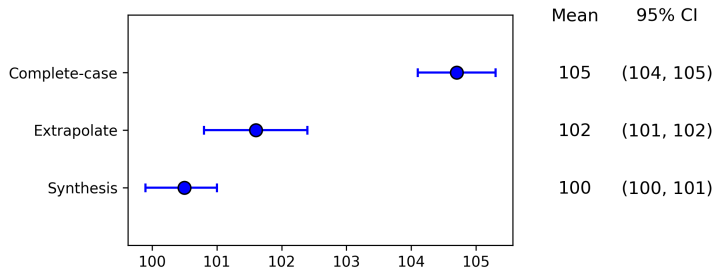
- Draw random SBP based on tables for each unit
 - Normal distribution by age, gender, height
- Compute mean then repeat

⁶Flynn et al. 2017 *Pediatrics* 140(3):e20171904

NHANES Data – Synthesis



Results – Mean SBP



Conclusion

Positivity is an important, but often unrecognized, assumption

- Appears with our exchangeability assumptions

Synthesis of statistical and mathematical models to fill in missing data with nonpositivity

Thank you



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Corresponding Publications

- Zivich et al. *Epidemiology* 2024;35(1):23-31
- Zivich et al. *JRSSA* 2025;188(1):158-180
- Zivich et al. *arXiv:2503.02789*

Appendix

Why Positivity

Intuition: need some with $X = x$ to have Y observed at least sometimes so can 'stand-in' for without Y measured

Formally: note that

$$E[Y \mid X = x, R = 1] = \frac{E[Y R I(X = x)]}{\Pr(R = 1 \mid X = x) \Pr(X = x)}$$

with positivity preventing division by zero

So, *positivity* comes along with *exchangeability*